

HING ONG

Curriculum Vitae
hing5ong5@gmail.com
<https://hingong.github.io/>

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EDUCATION

PhD	University at Albany, State University of NY, Atmospheric Sciences Dissertation: “The significance of the nontraditional Coriolis terms in tropical large-scale dynamics”	2020
MS	National Taiwan University, Atmospheric Sciences Thesis: “Effects of artificial local compensation of convective mass flux in the cumulus parameterization”	2016
BS	National Taiwan University, Atmospheric Sciences	2014

PUBLICATIONS

Peer-Reviewed Publications in Atmospheric Sciences

- 2026 Zhang, A., Yang, D., **Ong, H.**, & Tan, Z., Understanding the evolution of global atmospheric rivers with a vapor kinetic energy framework. *Geophys. Res. Lett.*, 53, e2025GL119841.
- 2025 **Ong, H.**, Scale analysis for the Madden–Julian oscillation. *Q. J. R. Meteorol. Soc.*, 151(772), e5028.
- 2025 **Ong, H.**, & Yang, D., Westward- or eastward-propagating Rossby waves: Schematic illustrations. *J. Atmos. Sci.*, 82(9), 1903–1910.
- 2024 **Ong, H.**, & Yang, D., Vapor kinetic energy for the detection and understanding of atmospheric rivers. *Nat. Commun.*, 15, 9428.
- 2022 **Ong, H.**, & Yang, D., The compressional beta effect and convective system propagation. *J. Atmos. Sci.*, 79(8), 2031–2040.
- 2021 Skamarock, W. C., **Ong, H.**, & Klemp, J. B., A fully compressible nonhydrostatic deep-atmosphere equations solver for MPAS. *Mon. Weather Rev.*, 149(2), 571–583.
- 2020 **Ong, H.**, Comments on “On the structure and formation of UTLS PV dipole/jetlets in tropical cyclones by convective momentum surges”. *Mon. Weather Rev.*, 148(11), 4693–4695.

- 2020 **Ong, H.**, & Roundy, P. E., The compressional beta effect: Analytical solution, numerical benchmark, and data analysis. *J. Atmos. Sci.*, 77(11), 3721–3732.
- 2020 **Ong, H.**, & Roundy, P. E., Nontraditional hypsometric equation. *Q. J. R. Meteorol. Soc.*, 146(727), 700–706.
- 2019 **Ong, H.**, & Roundy, P. E., Linear effects of nontraditional Coriolis terms on intertropical convergence zone forced large-scale flow. *Q. J. R. Meteorol. Soc.*, 145(723), 2445–2453.
- 2017 **Ong, H.**, Wu, C. M., & Kuo, H. C., Effects of artificial local compensation of convective mass flux in the cumulus parameterization. *J. Adv. Model. Earth Syst.*, 9(4), 1811–1827.

In-Progress Works in Atmospheric Sciences

- 2026 **Ong, H.**, The nontraditional Coriolis terms and trade-wind cumuli. Preprint. doi:10.22541/essoar.176945330.02654752/v1
- 2026 **Ong, H.**, Larson, V. E., The Coriolis effects in parameterization of atmospheric turbulence and convection. Preprint. doi:10.22541/essoar.177135891.11467584/v1
- 2026 **Ong, H.**, Jung, C., Wang, J., Kotamarthi, V. R., & Sever, G., Evaluation of near-surface air temperature and surface energy fluxes in a convection-permitting dynamical downscaling over the contiguous United States. Submitted.
- 2026 **Ong, H.**, Hughes, O., Herrington, A., Jablonowski, C., Lauritzen, P. H., & Yang, D., ITCZ and the nontraditional Coriolis terms. Abstract.

Peer-Reviewed Publication in Linguistics

- 2025 **Ong, H.**, Functional aspiration in Taiwanese. *Taiwan Journal of Linguistics*, 23(2), 51–81.

HONORS AND AWARDS

- 2020 **Climate and Global Change Postdoctoral Fellowship**, NOAA (declined)
- 2019 **Government Scholarship to Study Abroad**, Ministry of Education, Taiwan

RESEARCH EXPERIENCE

Independent Scholar

2025 to current

Coordinated cross-institutional model intercomparison project
 Brought CESM developers at NCAR into the project
 Brought E3SM developers at Univ. of Michigan into the project
 Brought CLUBB developers at Univ. of Wisconsin-Milwaukee into the project
 Framed scientific questions for the project

Convened regular meetings for the project
 Formulated a minimal model for the Madden–Julian oscillation
 Formulated the Coriolis effects in atmospheric turbulence parameterization
 Designed a model hierarchy for deep-atmosphere dynamical cores
 Designed a model hierarchy for turbulence parameterization with the Coriolis effects
 Adapted the dynamics of CLUBB atmospheric turbulence parameterization
 Adapted the model interface of CESM, MPAS-A, and CLUBB
 Conducted sensitivity simulations of Coriolis effects with CESM with MPAS-A
 Conducted sensitivity simulations of Coriolis effects with SAM
 Conducted sensitivity simulations of Coriolis effects with CLUBB

Postdoctoral Appointee, Argonne National Laboratory 2023 to 2025
 Performed climate dynamical downscaling with WRF model
 Evaluated the surface fluxes of the model against observations
 Conducted sensitivity simulations of land models in WRF
 Managed petabytes of data storage

Postdoctoral Scholar, University of California, Davis 2020 to 2023
 Adapted the dynamics of SAM atmospheric model.
 Performed spectral analysis to model simulation data.
 Formulated the prognostic equation of vapor kinetic energy.
 Analyzed MERRA2 and ERA5 reanalysis data.

Student Researcher, University at Albany, State University of NY 2017 to 2020
 Formulated a numerical idealized circulation model.
 Analyzed rawinsonde and ERA-Interim reanalysis data.
 Derived analytical equatorial wave solutions.
 Developed a benchmarking test for model dynamics.
 Adapted the dynamics of MPAS atmospheric model.

Research Assistant, National Taiwan University 2016 to 2017
 Participated in a scientific planning group in a field experiment.
 Composed a progress report.

Student Researcher, National Taiwan University 2014 to 2016
 Formulated a cumulus parameterization scheme.
 Adapted the dynamics and physics of WRF atmospheric model.

TEACHING EXPERIENCE

Adjunct Faculty, University of Maryland Global Campus 2026 to current
 Fundamentals of Physics I

STEM Teacher, Seattle Taiwanese Language Association 2023 to current
 Designed STEM activities for K-to-11 students in Taiwanese Taigi
 Delivered STEM activities at Taiwanese Language Summer Camps

Teaching Assistant, University at Albany, State University of NY Applications of Subseasonal to Seasonal Dynamics Ocean Science Water and Climate Change Atmospheric Dynamics	2018 to 2020
Teaching Assistant, National Taiwan University Lab. of Synoptic Meteorology (<i>de facto</i> Lecturer) Fluid Mechanics Program and Scientific Computing	2014 to 2016

PROFESSIONAL SERVICE

Peer-Reviewed Articles for:

Journal of Atmospheric Sciences
 Geophysical Research Letters
 Monthly Weather Review
 Journal of Geophysical Research: Atmospheres
 Journal of Climate

Coordinated Seminar Series for:

2022 Winter Atmospheric Science Seminar, University of California, Davis

Panel-Reviewed STEM Education Materials for:

教育部國家教育研究院「臺灣台語新詞、專有名詞及學術名詞對譯審譯組」
 (Taiwanese Taigi New Words, Proper Nouns, and Academic Nouns Translation Panel,
 National Academy for Educational Research, Ministry of Education, Taiwan)

INVITED LECTURES (SELECTED)

- 2025 “Revisiting tropical dynamics: New insights on the MJO and double-ITCZ bias”
 Physical Sciences Lab., NOAA, Boulder, CO, Jun 12.
 Dept. of Atmospheric and Climate Science, Univ. of Washington, Seattle, WA, Jun 23
- 2025 “ITCZ and the nontraditional Coriolis terms”
 CESM Workshop, Boulder, CO, Jun 9.
 Climate & Global Dynamics Lab., NCAR, Boulder, CO, Feb 5.
- 2024 “Pressure perturbation in mesoscale meteorology”
 Dept. of Geography and Meteorology, Valparaiso Univ., Valparaiso, IN, Mar 25.
- 2022 “Káng 風 soat 雨 òe 大氣” (Talk about wind, rain, and atmosphere)
 Sè-kài Tâi-oân Bûn-hòa Lûn-tôaⁿ (World Taiwanese Culture Forum), Online, Nov 12.
 Delivered in Taiwanese Taigi.

- 2021 “The nontraditional Coriolis terms and convective system propagation”
Geophysical Fluid Dynamics Lab., NOAA, Princeton, NJ, Sep 23.
- 2020 “The significance of the nontraditional Coriolis terms in tropical large-scale dynamics”
Research Center for Environmental Changes, Academia Sinica, Taipei, Taiwan, Jan 10.
Dept. of Atmospheric Sciences, National Taiwan Univ., Taipei, Taiwan, Jan 9.
- 2019 “The significance of the nontraditional Coriolis terms in tropical large-scale dynamics,”
Dept. of Earth, Atmospheric and Planetary Sciences, Massachusetts Institute of
Technology, Cambridge, MA, Oct 30.
Mesoscale and Microscale Meteorology Lab., NCAR, Boulder, CO, Jul 25.

LANGUAGES

English: Professionally proficient

Taiwanese Taigi: Native (my official name since Dec 2021, Hing Ong)

Chinese Mandarin: Native (my official name until Dec 2021, Heng Wang)

OUTSTANDING SKILLS

Model Formulation: using partial differential equations.

Model Development: using Fortran, Matlab, or Python

Data Analysis: using Fortran, Matlab, NCL, Python, or Grads

RESEARCH INTERESTS

Geophysical Fluid Dynamics

Earth System Modeling